10a) Internetworking with Routers in Cisco Packet Tracer

Aim:

To design and configure a simple internetwork using a router and two PCs.

Procedure:

Add one router and two PCs in cisco packet tracer.

Connect PC0 to Router 1 Fast Ethernet 0/0 and PC1 to Fast Ethernet 0/1 using copper straight - through cable.

Configure the router using CLI commands:

```
enable
configure terminal
interface Fast Ethernet 0/0
ip address 192.168.10.1 255.255.0
no shutdown
exit
interface Fast Ethernet 0/1
ip address 192.168.20.1 255.255.255.0
no shutdown
exit
```

subnet mask for both PCs: 255.255.255.0

IP Address Configuration

Device	IP Address	Default Gateway
PC0	192.168.10.2	192.168.10.1
PC1	192.168.20.2	192.168.20.1

Testing :

Use Add Simple PDU to send a packet from PC0 to PC1 and verify successful communication

Result :

Thus, a simple internetwork was configured successfully using a router and two PCs in Cisco Packet Tracer.

10/26 10b) Internetworking using wireless router, DHCP Server and internet cloud.

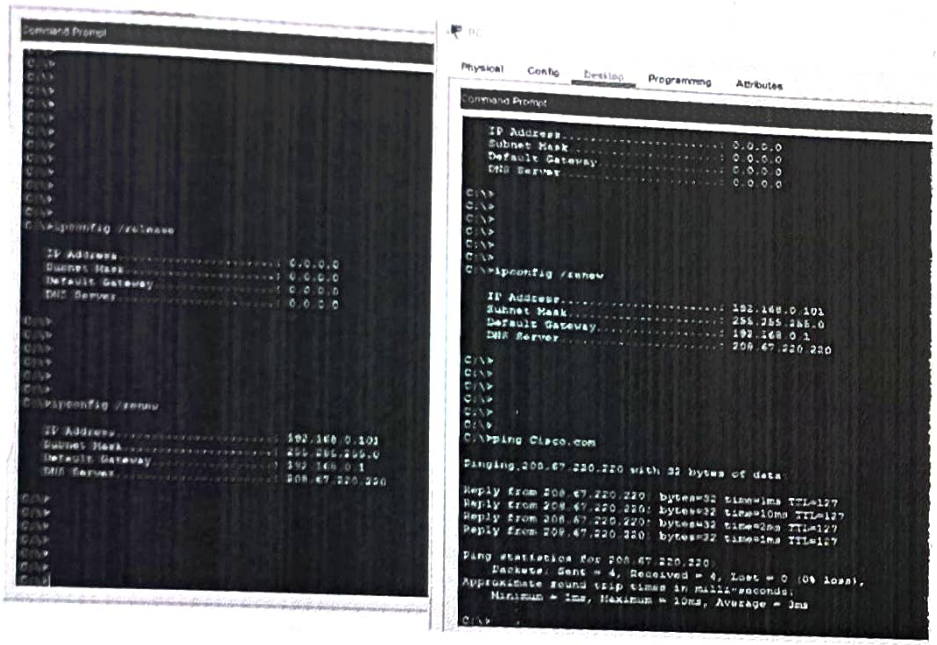
AIM:

To design and configure a network using a wireless router, DHCP server and internet cloud.

Procedure:

- Add a wireless router, PC, laptop, cable modem, internet cloud and cisco.com server.
- Connect the devices using suitable cable.
- Set the wireless router SSID to HomeNetwork and enable DHCP.
- Configure the laptop's wireless module and connect it to HomeNetwork.
- Set the PC to obtain an IP address automatically using DHCP.
- Configure the server IP as 208.67.220.220, DNS 208.67.220.220 and starting IP 208.67.220.1.
- Enable DNS service and create an A record for Cisco.com pointing to 208.67.220.220.

- Configure the server IP as 208.67.220.220 with subnet mask 255.255.255.0.
- Verify connectivity using ipconfig /renew and ping cisco.com.



Result:

Thus, a network using a wireless router, DHCP server and internet cloud was configured in Cisco Packet Tracer connectivity was tested successfully